

Credit Cards and Retail Firms: Historical Evidence from the U.S.

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Research Question: Have credit card networks benefited retail firms?

Roadmap

1. **Document Trends:** Banks have replaced retail firms as suppliers of short-term consumer credit in the U.S. since 1958
2. **Causal Identification:** Natural experiment shows that this increases firms' profits and relieves them of risk
3. **Mechanisms and Model:** Structural model with firm-level data to answer:
 - ▶ What determines firms' choice to accept cards?
 - ▶ How large is the net cost reduction?
 - ▶ How much do credit payment options affect consumer choice over retailers?

Main Finding: Firms accept bank cards to reduce costs by about 3%, which benefits both firms and consumers

Historical Context

“Most Americans had a dozen or more different forms of credit... an account at the neighborhood pharmacy, the neighborhood grocery store, and a half-dozen other local stores”

“It was the smaller merchants who first came around... The man almost kissed my feet... His accounts receivable were dragging him under”

“I looked at the size of his accounts: \$4.58. \$12.82.¹ He was sending out monthly bills. Then the customers paid him maybe three or four months later”

“The bank would guarantee him payment—within days instead of months—and would take over the role of collecting from the customers... [while] taking a 6 percent cut.”

-Bank of America salesman, 1958, on enrolling merchants (**nocera'piece'2013**)

¹Inflation since 1958 $\approx$ 10x

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“A dozen or more” in 1958 is down to 3.2 avg cards today, down from 4 in 1990 (SCF) despite a threefold increase in spending share on cards

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Why Might Firms Not Benefit?

Two Reasons:

- ▶ Bank, especially if monopoly, may be able to charge more than cost savings (**rochet platform 2003, edelman price 2015, wang regulating 2023**)
- ▶ Interoperable credit cards issued by banks may make firms better substitutes and force them to lower prices

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A (small) change in profits is equal to:

$$\underbrace{\frac{d\pi}{PQ}}_{\text{Change in Profit Rate}} = \underbrace{-(\sigma - 1)\mu \frac{dC}{C}}_{\text{Change due to Cost}} + \underbrace{d\mu}_{\text{Change in Markup}} \quad (1)$$

Where $P \equiv (1 + \mu) C$ and $\sigma \equiv -\frac{\partial Q}{\partial P} \frac{P}{Q}$

Roadmap

Section 1: Aggregate Trends

- ▶ Bank credit rose and displaced store credit
- ▶ Stores spend less on credit-related labor

Section 2: Identification of Real Effects

- ▶ An unexpected Supreme Court ruling in 1978 affected bank card supply differently across states
- ▶ Bank card usage rose, store credit fell, profitability increased, and firms already on bank card network lost sales

Section 3: Mechanisms, Structural Model

- ▶ Discrete-choice model of consumer choice over retailers
- ▶ Model allows credit to change **costs** and **markups** as free parameters
- ▶ Simulated Method of Moments to fit model to results from part (2)

Related Literature

Credit Expansions beck'finance'2008, beck'big'2010, chatterji'entrepreneurial'2012, cuesta'price'2021, kroszner'regulation'2014, jayaratne'entry'1998, kroszner'what'1999, chong'effects'1991, schularick'credit'2012, mian'what'2014, mian'how'2020, julia'fonseca'how'nodate, herkenhoff'impact'2021

- ▶ Formal credit may crowd out informal
- ▶ Case of tech-driven credit boom (this tech: verification at distance and scale)

Credit Bundling and Trade Credit banner'competition'1958, brennan'vendor'1988, benetton'credit'2022, low'auto'2020, murfin'who'2019

- ▶ Non-durable products, atomistic consumers (unlike B2B)
- ▶ Correlation between demand shocks and default suggests undiversified retailers should be particularly *poor* holders of own customers' default risk

IO of Payments rochet'platform'2003, edelman'price'2015, wang'regulating'2023, brunnermeier'mobile'2023, agarwal'real'2020, huynh'equilibrium'2022, arango'consumer'2015, evans'paying'2004

- ▶ Quantitative evidence of aggregate impact when technology can change costs

Preview of Main Findings

1. Store cards (non-interoperable, with firms bearing default risk) were 75% of credit card spending in 1970, but 10% in 2020.
2. More than half of bank credit growth caused by the end of usury (maximum interest rate) regulation in 1978 substituted for less retail lending.
3. In net, this increased firm profits and net entry of retail firms
4. A model estimates that this reduced retail industry's net cost by about 3%, dominating firms' lost markups from interoperability
 - ▶ The net effect of bank cards is positive for firm profits and consumer surplus
 - ▶ This is only true because banks are much more efficient than firms at credit supply, and the estimated importance of interoperability is small

Data Sources

- ▶ **Consumer Credit Use:** Survey of Consumer Finance
- ▶ **Firm Credit Supply and Profits:** Census QFR (industry-level), Compustat (firm-level)
- ▶ **Sales and Survival by Location:** Dun and Bradstreet (establishment-level)
- ▶ **Net Entry:** US Census County Business Patterns (establishment-level)
- ▶ **Usury Regulation:** “Cost of Personal Borrowing in the United States” (state-level)

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- ▶ **Bank Card Acceptance:** Yellow Pages (New establishment-level data)

New Data on Bank Card Acceptance

MASTER CHARGE

One Bill - One Check
Shopping convenience at
Thousands of Merchants in
this area.



Look for the Master Charge
Sign at the Merchants
Shown Below

BANKS
WEST SIDE BANK
MainAv&JacksnScrn--**346-8767**

ART GALLERIES & DEALERS
GALERIE des BEAUX ARTS
352AdmsAvScrn--**343-8591**

**AUTOMOBILE AIR
CONDITIONING EQUIPMENT**
**KAPLAN'S SERV AUTO
RADIATOR CO**
810WyoAvScrn--**344-3914**

**AUTOMOBILE BODY
REPAIRING & PAINTING**
**DERENICK'S REPR & BODY
SHOP** 460 N MainTay-----**562-1260**
**KAPLAN'S SERV AUTO
RADIATOR CO**
810WyoAvScrn--**344-3914**
TIBERI'S BODY SHOP
Rr708 N MainAvScrn--**344-5118**

Credit Card & Other Credit Plans

**PITTSBURGH NATIONAL
BANKAMERICARD**



CHARGE CARD

Lets You Do Your
Shopping At Merchants
Most Convenient To You.

"FOR INFORMATION CALL"
**BANKAMERICARD
MERCHANTS**
OAKMONT

Four The Girls Shop
434AlleghenyRiverBlvd
Oakmont--**828-4406**

Verona Auto Parts Co
827AlleghenyAvOakmont--**828-8500**

PENN HILLS
Associated Hardware
EastHillsShpgCtr--**731-4845**
Penn Hills Auto Parts
119PennoakDrPitsbrgh--**242-8626**
Penn Hills Electronic Service
1924UniversalRdPennHills--**793-3005**
Russ Plumbing
11752FrankstownRd--**242-0111**

STANTON HEIGHTS
Associated Hardware
StantonHtsShpgCtr--**661-2654**

Source: U.S. Telephone Directory ("USTD") Collection, Library of Congress

Data Collection

- ▶ **Step 1:** Identify 1,155 business directories since 1950 comprising 3,130 region-years from universe of Library of Congress Online directories
- ▶ **Step 2:** Identify 378 instances of explicit bank card acceptance listings comprising 25,519 firm-years
- ▶ **Step 3:** Link to Dun and Bradstreet establishment-level panel using phone numbers, addresses, and business names (2,956 unique firms; 81% of listings which overlap with D&B coverage)

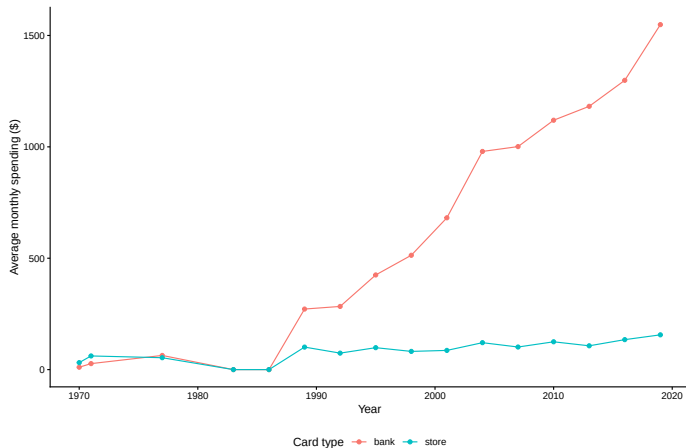
▶ Coverage in Time

▶ Coverage over Counties

Aggregate Trends

Bank Cards Displaced Store Credit

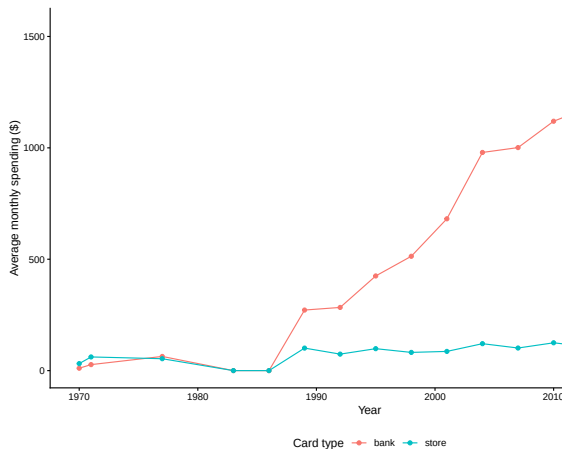
From the household perspective



Source: SCF

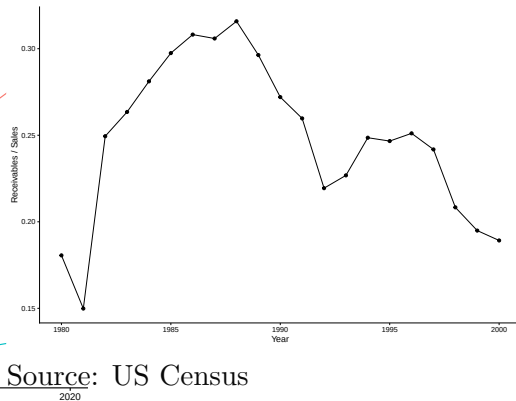
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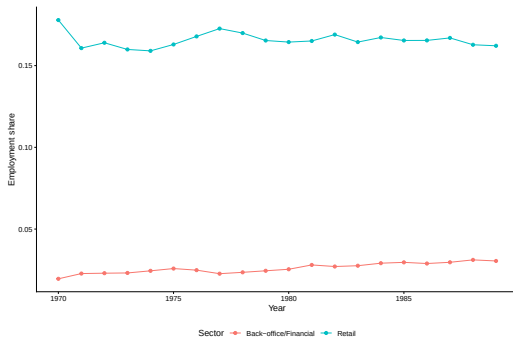
And from the firm perspective



Source: US Census

Labor Cost Reductions

Retail firms once spent 4% of their wage bill on “bookkeepers and bill collectors”



Source: US Census

- 78% drop since 1970, compared to 63% drop in wholesale sector

Identification: Abolition of Usury Limits

The Equilibrium Effect of Bank Cards

Goal: estimate the effect of bank credit card usage on firm outcomes

This requires variation in bank credit card supply that is:

- ▶ Is not confounded by other forms of credit (e.g. mortgages, commercial loans)
- ▶ Is not confounded by credit demand
- ▶ Varies across markets
- ▶ Is unanticipated

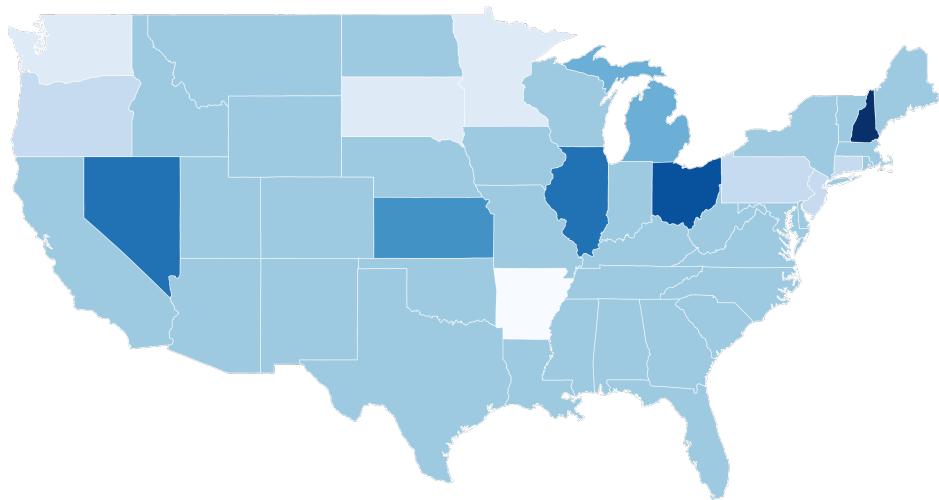
The 1978 *Marquette* decision by the U.S. Supreme Court creates such an ideal source of variation

1978 *Marquette* decision

- ▶ In 1978, supreme court ruled banks can lend across state lines at their home state's usury rate (maximum legal interest rate)
- ▶ Banks located in deregulated states expanded lending to consumers in regulated states
- ▶ Large banks such as Citibank and Bank of America migrated HQ's to no-limit states (South Dakota and Delaware, respectively) in 1981-1982
- ▶ Revolving credit (cards) regulated separately from installment loans – no effect on, e.g., mortgages and industrial loans
- ▶ Suggests a **single-treatment-time** Difference-in-Difference design with standard parallel trends assumption

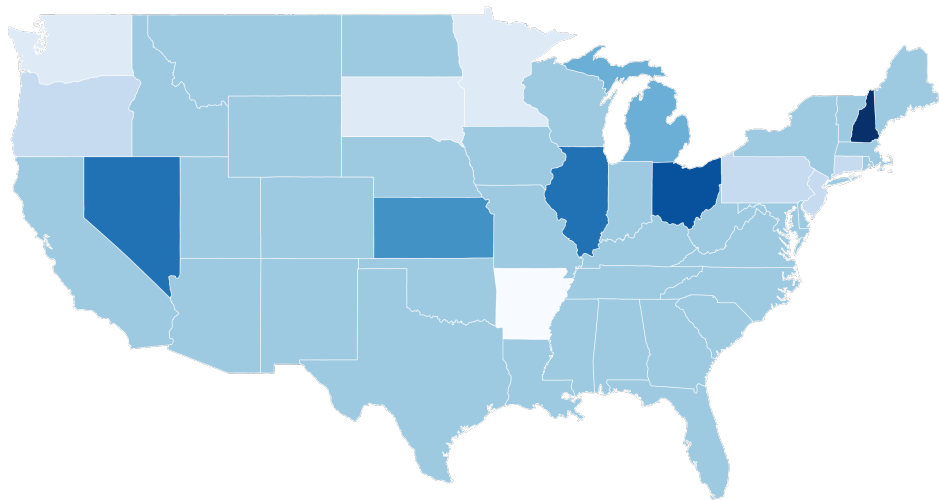
Support for Parallel Trends: Usury rates varied idiosyncratically in 1977

Usury Rate Limits, 1977

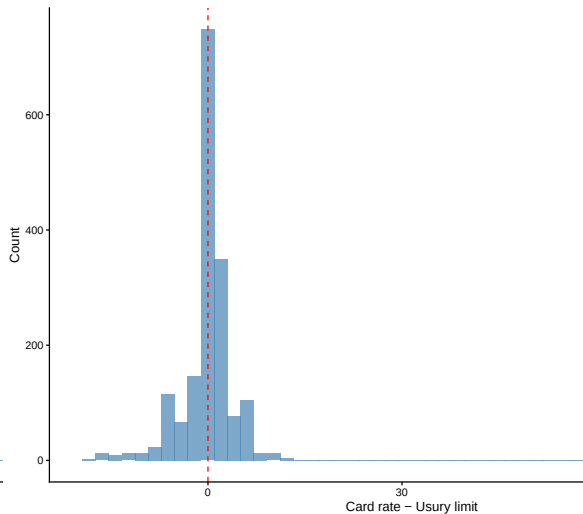
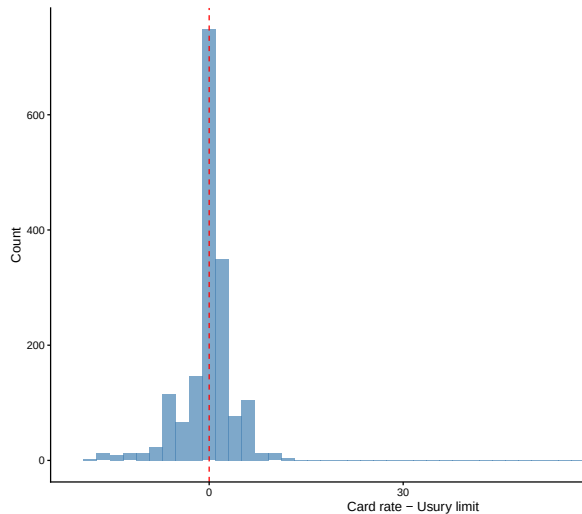


Support for Parallel Trends: Usury rates varied idiosyncratically in 1977

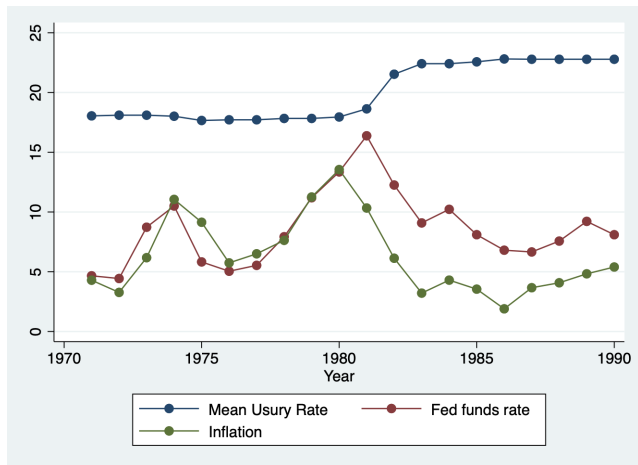
Usury Rate Limits, 1977



Relevance: usury regulation was tightly binding

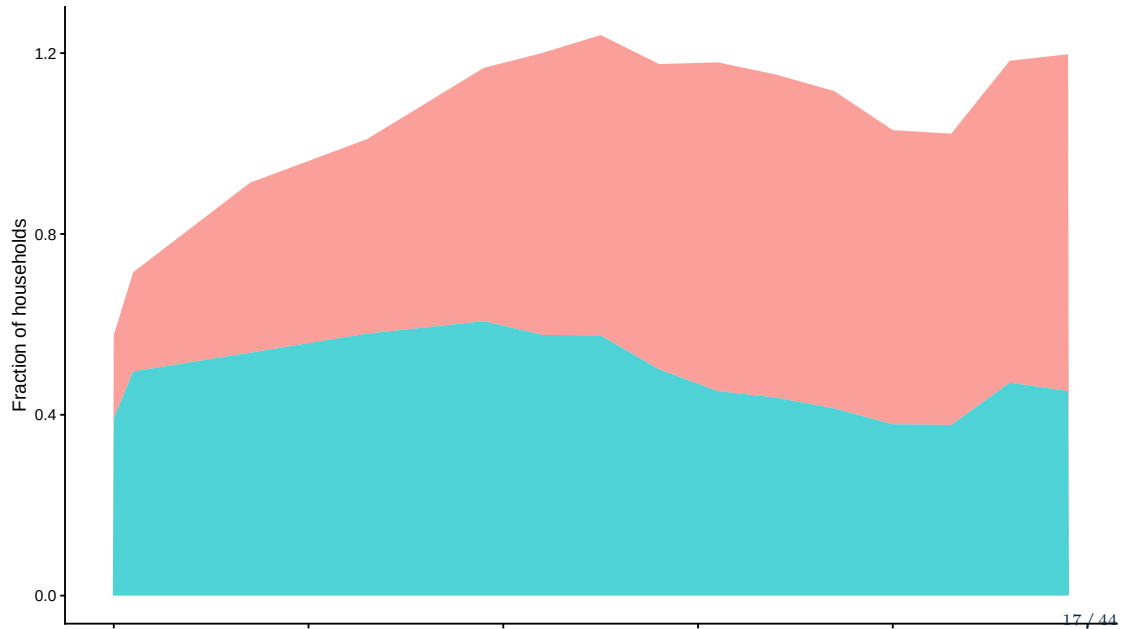


Relevance: usury regulation was tightly binding



- Usury lending rate was barely above inflation and fed funds rate (lower limit of bank marginal cost)

Growth in Aggregate Bank Card Lending



Main estimating equation

I use a dynamic differences-in-differences design:

$$Y_{it} = \alpha_i + \alpha_t + \beta_{t \neq 1977} T_i + \varepsilon_{it}$$

- **Comparison:** markets where customers suddenly had access to more bank cards vs. where bank cards had already been widely available

Main estimating equation

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$$Y_{it} = \alpha_i + \alpha_t + \beta_{t \neq 1977} T_i + \varepsilon_{it}$$

- ▶ **Comparison:** markets where customers suddenly had access to more bank cards vs. where bank cards had already been widely available
- ▶ Y_{it} are various outcomes of interest
- ▶ i are states/counties/firms (data permitting), t are years
- ▶ T_i is 1 if the state had a usury limit below 18% in 1978
 - ▶ **zinman impact 2003** finds a 5.5 pp increase in bank card holding in affected states from 1977 to 1983 using this treatment definition
 - ▶ 9 “treated” states representing 37 million Americans (16% of nation)
- ▶ Single treatment time means traditional two-way fixed effects DiD is not biased by time-varying treatment effects
- ▶ Std. errors clustered by state (**abadie when 2017**)

Receivables of Multi-Establishment Firm

Receivables are tracked at the firm-level, but many firms sell in many states

I calculate the firm-level treatment of firm i which has sales SLS_e establishments e in states $s(e)$ as a weighted average:

$$T_i = \sum_e \frac{T_{s(e)} \times SLS_e}{\sum_e SLS_e}$$

Retailers Reduced Lending



Data: All 185 retail firms covered by Compustat in 1978

Accounting Profits Increased

Dependent Variable:	rec_to_rev
Model:	(1)
<i>Variables</i>	
treated_post	-0.0100 (0.0166)
<i>Fixed-effects</i>	
year	Yes
gvkey	Yes
<i>Fit statistics</i>	
Observations	2,000
R ²	0.85525
Within R ²	0.00036
<i>Clustered (gvkey) standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

Triple Differences by Use of Cards for Transactions

Regression on accounting net income suffers several deficiencies:

- ▶ “Consumer” credit cards may allow entrepreneurs to start businesses, or increase demand (**mian·how·2020**)
- ▶ GAAP Net Income may not equal economic profits
- ▶ Few firms with public accounting data in 1977

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I address these with a triple-difference specification:

$$Y_{it} = \alpha_i + \alpha_t + \beta_{1,t \neq 1977} T_i + \beta_{2,t \neq 1977} T_i \textit{Clothing_Industry}_i + \varepsilon_{it}$$

Using establishment counts from Census CBP as Y_{it}

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Using establishment counts from Census CBP as Y_{it}

Under what assumption is β_2 a better estimate?

- ▶ Credit cards are equally useful to entrepreneurs in all industries
- ▶ Credit cards are more useful for payments in the clothing industry (empirically true)
- ▶ Net entry is a good proxy for economic profits

Bank Card Acceptance Across Industries in 1970

10Table: Credit Card Acceptance
by Industry, Scranton 1971

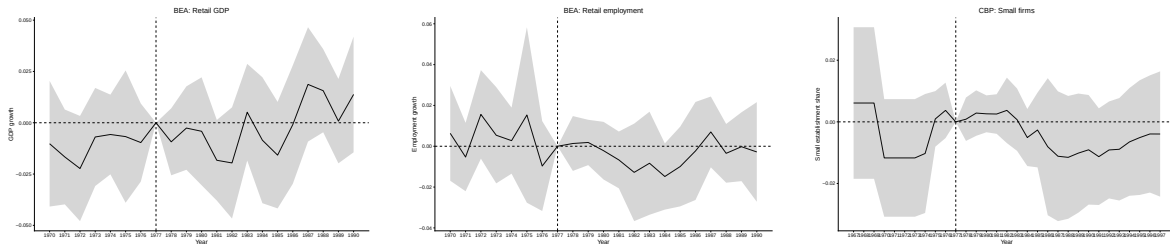
industry	acceptance	n
Food	6%	150
Auto	13%	110
Clothing	100%	95
Hardware	10%	30
Misc.	0%	177
Furnishing	0%	63
Restaurants	0%	200
Department Stores	0%	48

10Table: Credit Card Ownership
Over Time

year	Bank card	Store card	Any card
1970	0.18	0.39	0.57
1977	0.38	0.54	0.92
1983	0.43	0.58	0.91
2019	0.75	0.45	0.90

Cross-industry heterogeneity in card usage from new firm side data (left) aligns

Net Entry of Clothing Retail Firms



Data Source: County Business Patterns at County-Sic Code-Year level.

- Increase due to small firms – consistent with economies of scale in the provision of store credit

Effect on Sales and Survival

- ▶ Which firms are early adopters of bank credit cards?
- ▶ How do existing card-accepting businesses respond to greater use of the network?
- ▶ Card networks advertise increased sales for merchants
- ▶ Analyze D&B Establishment-level sales panel
 - ▶ Each market is a city x 4-digit-SIC
 - ▶ Regress growth rates from 1977-1983 on treatment x pre-acceptance
 - ▶ Survival times also a proxy for profitability

Which Firms Adopt Cards?

Dependent Variable:	acceptance	
Model:	(1)	(2)
<i>Variables</i>		
log_sls	0.0447*** (0.0127)	0.0448** (0.0129)
single	0.0434* (0.0187)	0.0441** (0.0181)
age		-0.0002 (0.0006)
<i>Fixed-effects</i>		
fe	Yes	Yes
<i>Fit statistics</i>		
Observations	1,970	1,970
R ²	0.25924	0.25932
Within R ²	0.02662	0.02673

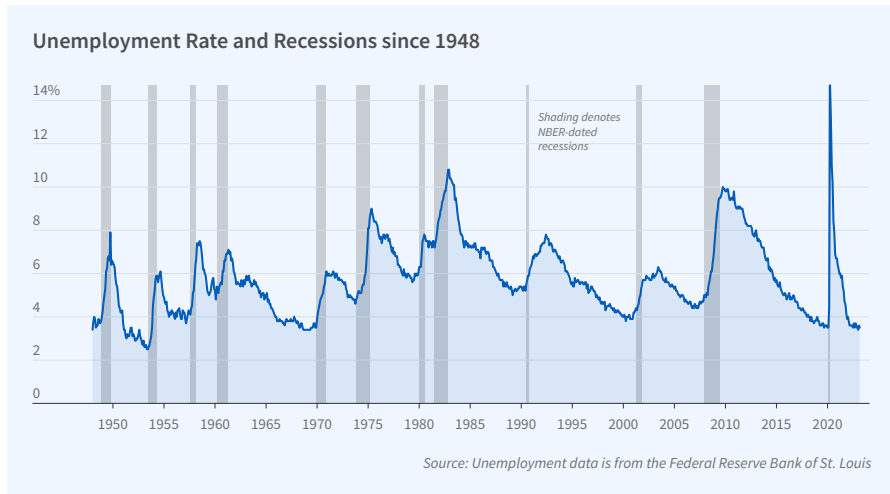
Clustered (industry) standard-errors in parentheses

How do Card-Accepting Businesses Respond?

Dependent Variables: Model:	SLS_growth (1)	d (2)	SLS_growth (3)	SLS_growth (4)
<i>Variables</i>				
acceptance	0.0188*** (0.0002)	-0.0194 (0.1885)	0.0007 (0.0028)	0.0193*** (0.0013)
log(SLS0+1)	-0.0037 (0.0057)	0.1777** (0.0346)	-0.0038 (0.0056)	-0.0039 (0.0057)
acceptance $\times$ treated	-0.0231** (0.0044)	0.8225** (0.1604)		-0.0237*** (0.0036)
treated			-0.0074** (0.0013)	-0.0042** (0.0010)
<i>Fixed-effects</i>				
STATE	Yes	Yes		
SIC1	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	525	525	525	525
R ²	0.00540	0.10533	0.00613	0.00607

Firm Resiliency to Risk

1975 and 1983 major spikes in unemployment before and after *Marquette*:



Firm Survival During Recessions

Dependent Variable: Model:	(1)	(2)	exit_rate (3)	(4)
<i>Variables</i>				
mean_log_sls	-0.0092*** (0.0004)	-0.0127*** (0.0008)	-0.0107*** (0.0005)	-0.0155*** (0.0008)
treated_single	-0.0006 (0.0027)	-0.0016 (0.0026)	-0.0020 (0.0015)	-0.0041** (0.0019)
single_recession	0.0459*** (0.0039)	0.0514*** (0.0046)	0.0305*** (0.0033)	0.0159*** (0.0022)
treated_recession_single	-0.0159* (0.0081)	-0.0204** (0.0099)	0.0003 (0.0113)	0.0072 (0.0084)
<i>Fixed-effects</i>				
fe	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	73,036	27,252	73,036	27,252
R ²	0.99726	0.99720	0.99718	0.99704
Within R ²	0.11503	0.17114	0.08924	0.12386

Clustered (STATE) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

- Treated and untreated states respond similarly to 1975 recession
- In 1983, small firms more likely to survive in treated states

Mechanisms and Model

Purpose of the Model

Goal: Quantitatively assess the relative importance of two effects of bank credit cards on retail firms:

- ▶ Cost Reduction
- ▶ Markup Reduction due to Interoperability

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Recall Equation (1):

$$\underbrace{\frac{d\pi}{PQ}}_{\text{Change in Profit Rate}} = \underbrace{-(\sigma - 1)\mu \frac{dC}{C}}_{\text{Change due to Cost}} + \underbrace{d\mu}_{\text{Change in Markup}} \quad (2)$$

I will impose functional form assumptions on supply and demand to structurally estimate this equation

Why a Discrete Choice Model?

The **berry·estimating·1994** family of models are designed to predict how prices and quantities respond to changes in competition (**einav·empirical·2009**) because they:

- ▶ Match the data with as few parameters as needed
- ▶ Are micro-founded by choices of heterogeneous, atomistic consumers
- ▶ Are widely used in antitrust (e.g. Nevo (2017))
- ▶ Come with a large established literature on what approximate magnitudes of parameters are reasonable

In this case, how are market-level prices and quantities affected by changes in (1) marginal cost and (2) product differentiation due to interoperable credit cards?

Intuition

Mechanics:

- ▶ Cost reduction helps those firms who accept bank cards, and benefits consumers on net
- ▶ Interoperability forces firms to reduce markups, benefiting customers at firms' expense

Preview of Results from the Model:

- ▶ Bank credit cards reduce net costs by 3%
 - ▶ Cost reduction is jointly identified with quality improvement
- ▶ Only 9bps reduction in markups is required to match the data
- ▶ Implied industry-wide average price reduction of 66bp

Structure of the Game

1. Take network structure (bank card acceptance) as exogenous
 - ▶ See **wang'regulating'2023** for a microfoundation of this process
2. Discrete firms simultaneously choose prices to optimize profits
 - ▶ **berry'estimating'1994** conditions guarantee unique equilibrium
3. Atomistic consumers, heterogeneous in taste for retailers and credit cards, choose retailers
 - ▶ Firms accepting an interoperable credit card are “nested” as in **miller'notes'2019**, **atalay'scalable'2023** and must lower markups to compete with each other

Demand

Consumer Utility: Consumer i selects from among firms j :

$$u_{ij} = \alpha p_j + \xi_j + \varsigma(g(j)) + (1 - \sigma) \varepsilon_{ij}$$

Demand

Consumer Utility: Consumer i selects from among firms j :

$$u_{ij} = \underbrace{\alpha p_j}_{\text{Price Sensitivity}} + \xi_j + \varsigma(i, g(j)) + (1 - \sigma) \varepsilon_{ij}$$

- p_j set optimally by each firm (more details later)

Demand

Consumer Utility: Consumer i selects from among firms j :

$$u_{ij} = \alpha p_j + \underbrace{\xi_j}_{\text{Quality}} + \varsigma(i, g(j)) + (1 - \sigma) \varepsilon_{ij}$$

- ▶ Free parameters to match any initial observed market shares
- ▶ Held constant in counterfactuals
- ▶ Changing cost cannot be separately identified from changing quality

Demand

Consumer Utility: Consumer i selects from among firms j :

$$u_{ij} = \alpha p_j + \xi_j + \underbrace{\varsigma(i, g(j)) + (1 - \sigma) \varepsilon_{ij}}_{\text{Consumer Taste for Firm}}$$

- ▶ Distributed Type-1 Extreme Value
- ▶ Includes consumer taste for credit product $\varsigma(i, g(j))$, and consumer taste for final product ε_{ij}
- ▶ Distributions of ε , $\varsigma + \varepsilon$ held constant in counterfactuals as g changes

Demand

Consumer Utility: Consumer i selects from among firms j :

$$u_{ij} = \alpha p_j + \xi_j + \underbrace{\varsigma(i, g(j))}_{\text{Consumer Taste for Credit Product}} + (1 - \sigma) \varepsilon_{ij}$$

- ▶ $g(j)$ is either 1 or 0 – accepts bank card or not
- ▶ Free Parameter $\sigma \in (0, 1)$ is the importance of credit cards for steering demand. $\sigma = 0$ means credit cards do not affect consumer choice
- ▶ Two firms with $g = 1$ must compete over same segment of consumers

Demand (Cont.)

Define “mean utility” $\delta_j = \alpha p_j + \xi_j$

Market shares of each firm are conveniently equal to:

$$s_j = \frac{\exp\left(\frac{\delta_j}{1-\sigma}\right)}{D_g^\sigma \sum_g D_g^{1-\sigma}}$$

Where D_g value for each network g is equal to

$$D_g = \sum_{k \in g} \exp \frac{\delta_k}{1-\sigma}$$

Supply

N firms compete over prices, given $g(j)$

Firm j 's First Order Condition:

$$(p_j - c_j)^{-1} = \frac{\partial s_j}{\partial p_j} = \frac{\alpha}{1 - \sigma} s_j (1 - \sigma \bar{s}_{j|g} - (1 - \sigma) s_j)$$

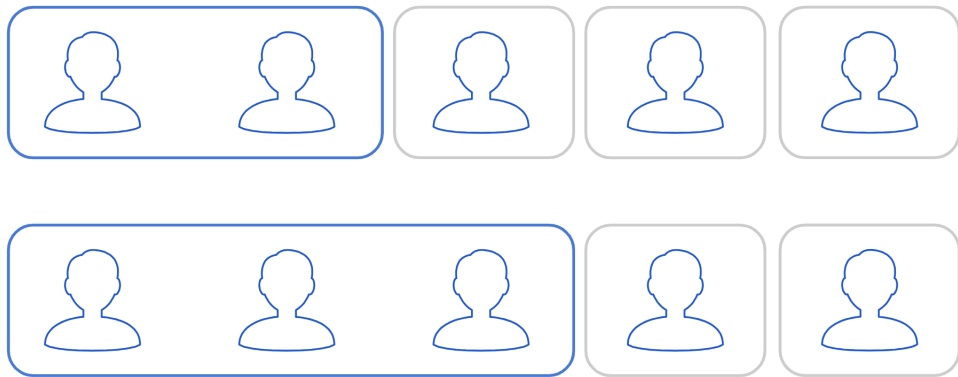
- ▶ A fraction $f(g)$ of consumers use credit card network g
- ▶ The firms' cost is $1 + \zeta f(g(j))$, $\zeta < 0$
- ▶ ζ is a free parameter chosen to fit the data
- ▶ The cost index of the industry at $g = 0$ is the numeraire

Estimating the Model

- ▶ Use data from 252 markets where Yellow Pages data and Dun & Bradstreet can be well matched
- ▶ Match the median market structure in terms of number of competing firms, realistic heterogeneity in market shares, and bank card acceptance
- ▶ Estimate σ and ζ via Simulated Method of Moments by matching the observed responses of store credit, sales, and heterogeneity by card acceptance from the *Marquette* Shock
- ▶ Standard procedure (**miller·notes·2019**) modified to match data on Sales (Price x Quantity) instead of Market Shares (Quantity)

Simulating Change in Network Structure

Solve for equilibria pre- and post-*Marquette* when new firms are added to credit card network:



Illustrative example with 5 firms, where network expands from 2 firms to 3 firms

Estimation Results

Parameters		
Parameter	Value	Method
Price Sensitivity α	-4	Atalay et al (2023)
Card Taste ς	0.07	Moments
Cost Change ζ	-0.07	Moments
N Firms	7	Observed
N Bank Networks	1	Observed
Top Firm Share	47%	Observed
Firm Store Card Share	83%	Observed
Consumer Bank Card Share	37%	Observed
Target Moments (My Estimates)		
Moment	Model	Data
Δ Store Card Share	-20%	-20%
Δ Bank Card Share	+5.5%	+5.5%
$\Delta SLS_{Notaccept}$	+27bp	+27bp
ΔSLS_{Accept}	-4.8%	-4.8%

Decomposing the Effect of Bank Cards

Variable	Baseline	Treated (Marquette)	No Cost Changes	Only Cost Changes
	(1)	(2)	(3)	(4)
Price Index	1.24	-66bp	-21bp	-45bp
Cost Index	0.99	-57bp	0	-57bp
Markup	0.25	-9bp	-21bp	+12bp

Implications for Regulators and Future Avenues for Research

UK Draft Legislation on BNPL

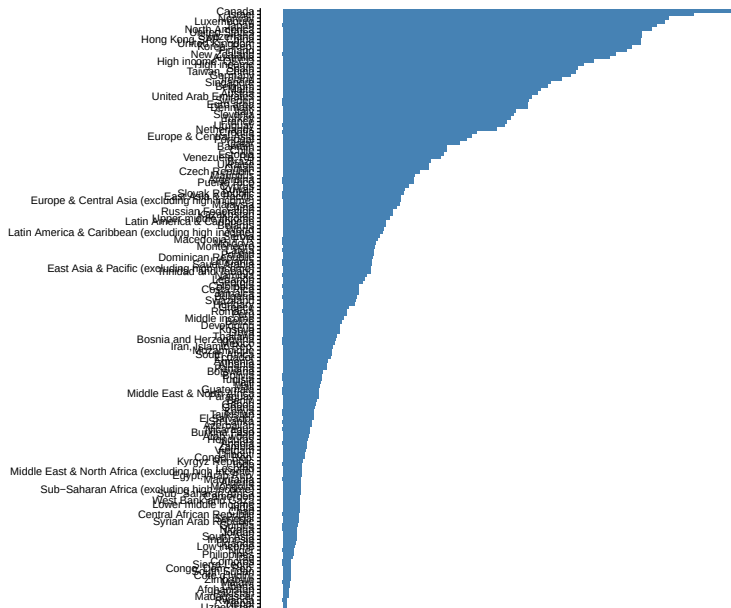
UK draft legislation includes specific carve-out for merchant-funded credit:

- Specifically, “borrower-lender-supplier agreements” for fixed-sum credit to individuals or relevant recipients of credit (which would include some small businesses) will become regulated where they are: (i) interest-free and repayable in 12 or fewer instalments within 12 months or less; (ii) provided by a person who is not the provider of goods or services that the agreement finances (*i.e.*, a third party and not the merchant); and (iii) is not otherwise exempt.

(toms.uk)

Why force default risk to be held by retailers?

Developing Countries Still Rely on Store Lending



Conclusion

- ▶ Nonfinancial retail merchants are a significant part of the consumer credit system and engage in “regulatory arbitrage”
- ▶ A monopolistic card network may still benefit merchants if it can better allocate real risk and real cost
- ▶ More research is needed on the outcomes of consumers who borrow from nonfinancial lenders, especially in developing countries

Thank You!



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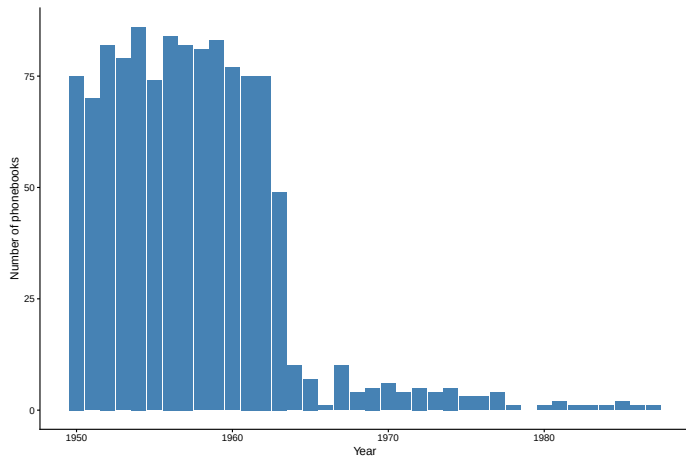
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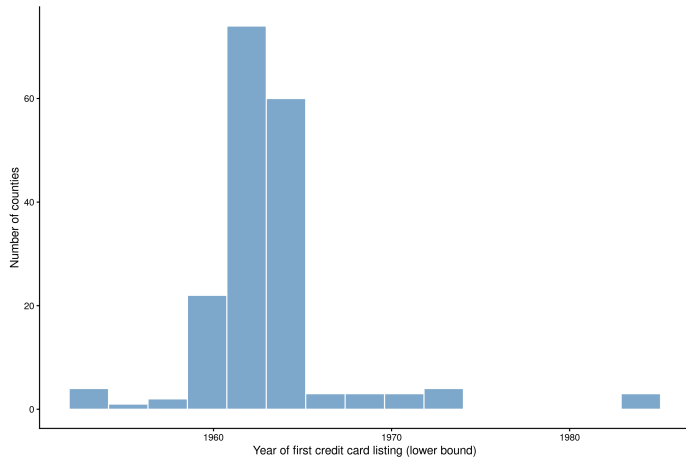


USTD: Coverage Over Time



Number of Directories by Year in USTD Universe [▶ Back](#)

USTD: Geographic Coverage



Dates of Bank Card Introduction [► Back](#)